ARITHMETIC TOPOLOGY, EXECUTABLE

primeknots

Named theorems of number theory, verified from scratch in C++ — where the process makes fitting the answer impossible.

github.com/DerekMarshall/primeknots · 63 tests, 0 failing · MIT

ACT I — THE FAILURE MODE

A survey that read as mathematics

- A research roadmap mixed real arithmetic topology with a fabricated object — a "chaotic operator O_x" with invented properties.
- In prose, the fiction was indistinguishable from the facts: same register, same citations-shaped scaffolding.
- Nothing in the reading experience separated the true claims from the false one.

The behavior, not the tool: **unverified synthesis** — assembling plausible text without a mechanism that could reject a wrong piece.

ACT I — THE FAILURE MODE

Why it survived five revisions

- Each critique was absorbed as camouflage: the text got more plausible, not more true.
- Every review pass raised the polish and lowered the visibility of the seam.
- Agreement with a reviewer is not evidence. Neither is agreement with expectation.

Related behaviors seen elsewhere in the project's own history: **fitting a result to an expected value**, and **fabricating a citation**. Same root — no falsifying step.

ACT I — THE MORAL

Rule 1

Agreement with expectation is evidence only when the process makes fitting impossible.

Never fit conventions to expected answers. If a published anchor comes out wrong, the bug is in the code or in our *reading of the source* — go fix the reading. Flipping a sign until the known number appears is falsification of the experiment.

Everything after this slide is the machine that enforces Rule 1.

ACT II

The dictionary and the ladder

One analogy — primes as knots — climbed one theorem at a time, each rung a running test.

ACT II — THE DICTIONARY

Primes ⇔ knots

- Spec $\mathbb{Z}[1/S]$ ↔ a 3-manifold with boundary (one "knot" per prime in S).
- Legendre symbol ↔ linking number.
- Rédei symbol ↔ triple (Milnor) linking.
- class group ↔ H_1; ζ-zeros ↔ a spectrum.

Morishita's *Knots and Primes* [M12] is the dictionary; the whole ladder is this one analogy, made executable.

```
Stage 0 reciprocity
Stage 1 linking matrix
Stage 2 Rédei / Borromean
Stage 3 4-rank
Stage 4 Chern–Simons (mod
2)
Stage 5 explicit formula
Stage 6 S₃ Dijkgraaf–
Witten
```

ACT II — STAGE 0

Reciprocity, the base case

- Legendre/Jacobi symbols, Tonelli–Shanks, Miller–Rabin — from scratch, u128 built-ins only.
- **Twin:** every symbol computed by Euler's criterion *and* by the reciprocity law — two algorithms, cross-checked.
- **Three coverages, stated separately** (not one "exhaustive to 10^7"):
  - the *reciprocity law* over all pairs of odd primes < 3×10^4 — 5,260,146 pairs;
  - both *supplements* over all 664,578 odd primes < 10^7;
  - the Euler-vs-reciprocity *twin* — 53,165,634 symbol agreements, exhaustive to 10^7.

```
✓ verified · reciprocity 5,260,146 pairs to 3×10⁴ · supplements 664,578
primes to 10⁷ · twin 53,165,634 agreements to 10⁷
```

ACT II — STAGE 1

The linking matrix

The linking matrix of primes

Stage 1 — $(p/q) = (-1)^{lk_2(p,q)}$ over primes ≡ 1 (mod 4). Renders emitted JSON only.

Linking heatmap

linked (symbol −1) unlinked (symbol +1)

The matrix $(p/q) = (-1)^{lk_2(p,q)}$ over primes ≡ 1 (mod 4). It is **symmetric** — the picture *is* quadratic reciprocity.

✓ verified · symmetry over 39175×39175 (1.53×10^9 bits, no sampling)

ACT II — STAGE 2

The Rédei symbol

- The triple symbol `[a,b,c] ∈ {±1}` is the mod-2 triple **Milnor invariant** — a Massey product in Galois cohomology (Morishita [M04]).
- **Rédei reciprocity:** full S_3 symmetry under all six permutations of the arguments — 2022 mathematics (Stevenhagen [S22]), verified over every valid triple in range.
- The 2-adic normalization is the delicate part: pinned to the source before any code (stage2-pinning §8), with *invariance senior to anchor*.

```
✓ verified · 702,788 triples × 6 = 4,216,728 symbol evaluations agree
```

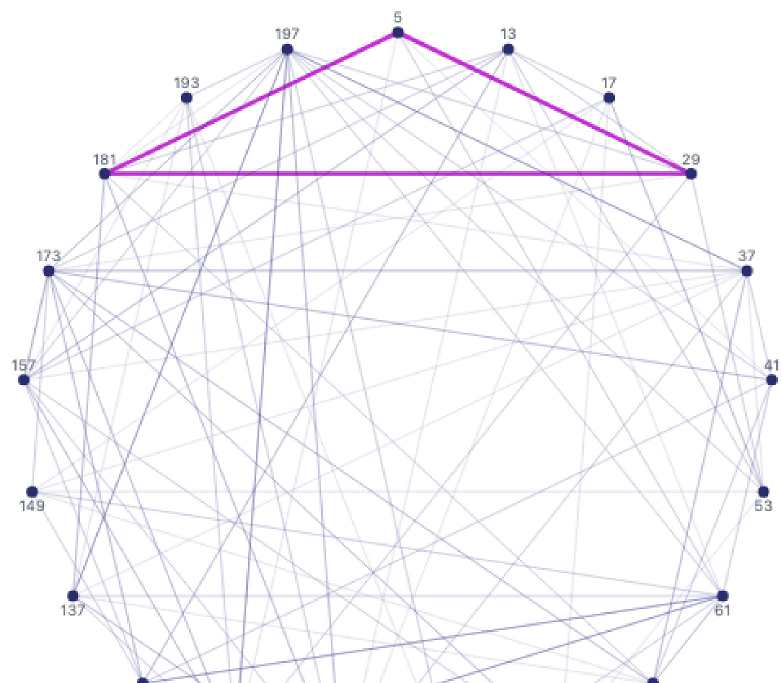
ACT II — STAGE 2

Borromean primes

Borromean prime triples

Stage 2 — pairwise-unlinked triples of primes ≡ 1 (mod 4) with Rédei symbol −1 (triple-linked). Renders emitted JSON only.

3-uniform hypergraph



Triples pairwise *unlinked* yet triple-*linked* (`[a,b,c]=−1`). Anchor: `[13,61,937] = −1`, the standard Borromean triple (Morishita [M12]); the *normalization* is pinned to Stevenhagen [S22]/Corsman [C07].

✓ verified · [13,61,937] = −1, the [M12] standard Borromean triple

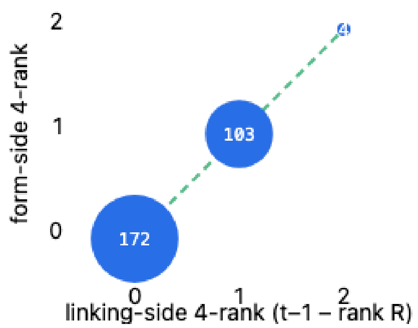
ACT II — STAGE 3

Class groups as H_1 — the 4-rank

Class groups and the Rédei–Reichardt 4-rank

Stage 3 — narrow class group $Cl^+(D)$ computed two independent ways. The scatter must be a perfect diagonal. Renders emitted JSON only.

4-rank: form composition vs. linking matrix



each cell = # discriminants with (linking corank t−1−rank R, form-side 4-rank); off-diagonal (red) = a Rédei–Reichardt disagreement

Aggregate

```
Rédei–Reichardt over 279 discriminants (primes ≡ 1 mod 4, t ≤ 4, D ≤ 20000):
form-side 4-rank == linking-side 4-rank: ALL MATCH ✓
```

Rédei–Reichardt: the 4-rank of the narrow class group (**form** side) equals the F_2-rank of the linking matrix (**linking** side). Two computations, one diagonal.

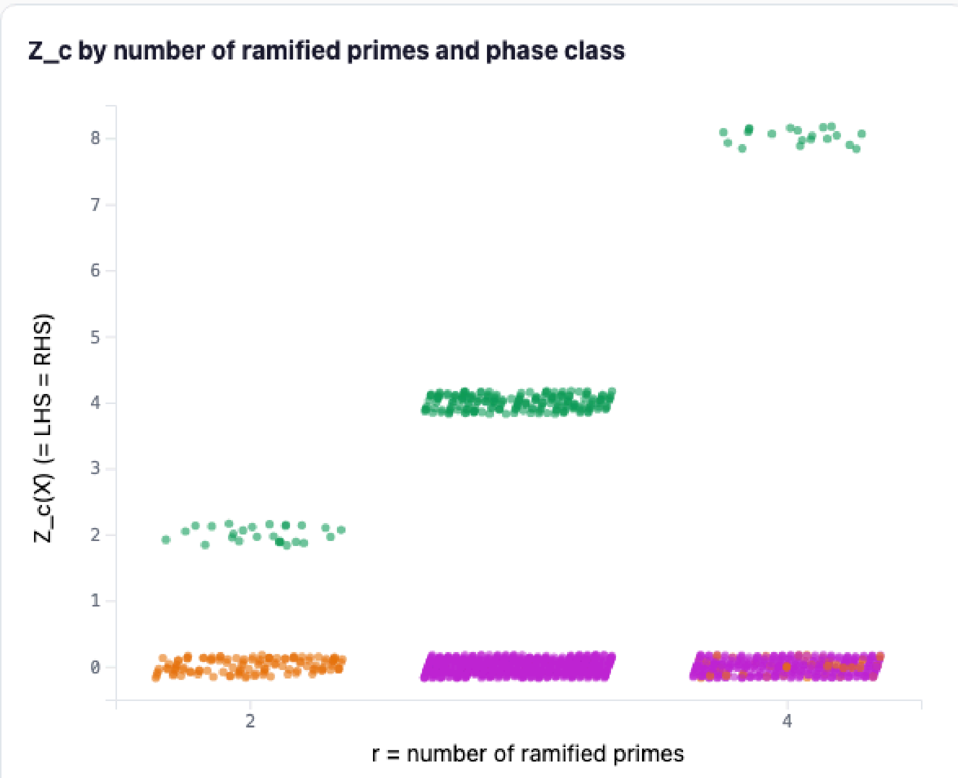
✓ verified · 1,986 discriminants; form-rank = linking-rank; oracle bnfnarrow

ACT II — STAGE 4

Arithmetic Chern–Simons (mod 2)

The arithmetic Chern–Simons partition function

Stage 4 — mod-2 Dijkgraaf–Witten Z_c(X) for $K=\mathbb{Q}(\sqrt{}(p_1\cdots p\_r))$, $p_i\equiv 1$ mod 4 (Hirano route). LHS (brute character sum) vs RHS (closed form). Renders emitted JSON only.



The mod-2 Dijkgraaf–Witten partition function Z_c, computed as a brute character sum (LHS) and a closed form (RHS) — in **separate translation units, disjoint authoring sessions**. Agreed on first comparison.

✓ verified · LHS = RHS over 386 gated prime sets (4,810 extended)

ACT II — STAGE 4 · HONESTY

The degenerate case, named as such

The mod-2 (Hirano) case is the degenerate one: Chern–Simons is $\mathbb{F}_2$-linear, so the identity *reduces to linear algebra on the linking matrix*.

- The contribution was the **pinning** and a **hypothesis erratum**: RESEARCH.md's "p = 2 over $\mathbb{Q}$" is unlicensed — CKKPPY Thm 1.1 needs odd p over a totally imaginary field.
- The odd-p Gauss-sum case is future work, not a result here.

A slide that oversold this would fail our own review.

ACT II — STAGE 5

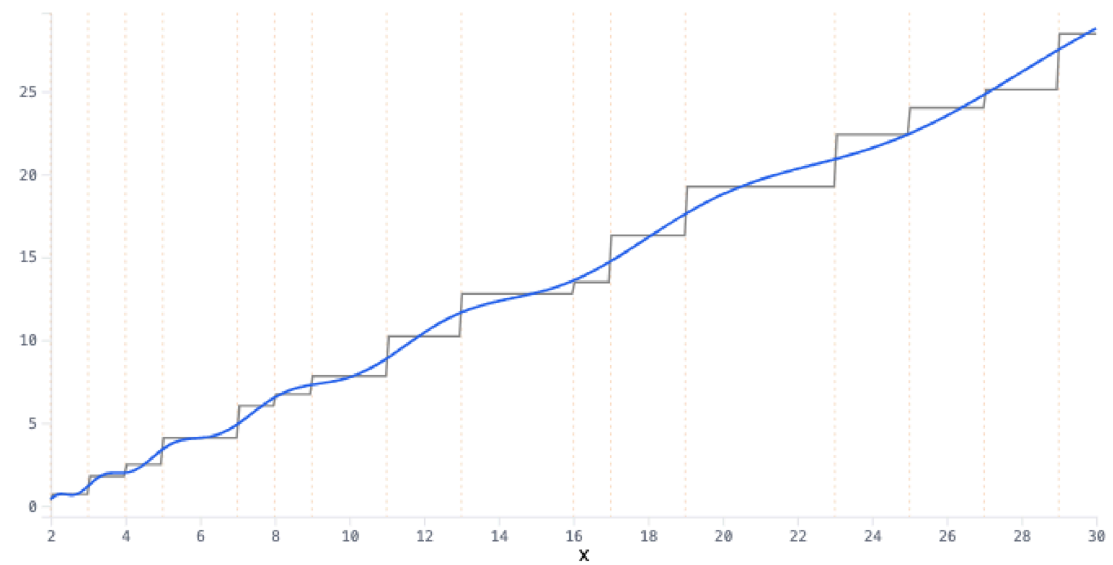
The Weil explicit formula

The explicit formula, and the Deninger gap made visible

Stage 5 — primes ⇄ zeta zeros (Weil/von Mangoldt explicit formula), beside a fully-constructed dynamical system where the same identity is geometry. Renders emitted JSON only.

$\psi_0(x)$ reconstructed from ζ zeros — the spectral side building the prime staircase

▶ play   m = **1** zeros   max err (away from jumps) = **1.88e+0**



■ ψ_0 reference (midpoint-normalized staircase) ■ Σ over first m zeros ■ prime-power jumps

The series converges to $\psi_0 = \frac{1}{2}(\psi(x^-)+\psi(x^+))$; at a prime power that is the jump midpoint

ψ(x) rebuilt from ζ zeros. Zeros by Riemann–Siegel, twinned against Euler–Maclaurin, completeness **Turing-certified** (not Gram's law). Beside it: a model flow (next slide).

✓ verified · first 10^4 zeros match Odlyzko to ≥8 dp (max dev 2.9×10^{-9})

✓ verified · converges to the jump midpoint ψ_0, not $\psi(x)$

ACT II — STAGE 5 · THE MODEL FLOW

Where the identity IS geometry — and where it isn't

- A subshift's Ruelle zeta: **Euler product over closed orbits = 1/det(I−L_s)**, exactly — orbits genuinely exist, the identity is geometry.
- Arithmetic satisfies the *same identity shape* (primes ↔ orbits, zeros ↔ spectrum)…
- …but **no such dynamical system is constructed**.

Deninger's foliated flow: not constructed — that gap *is* the program [D98].

✓ verified · Euler product = determinant, agreement ~9×10^{-16} over an s-grid

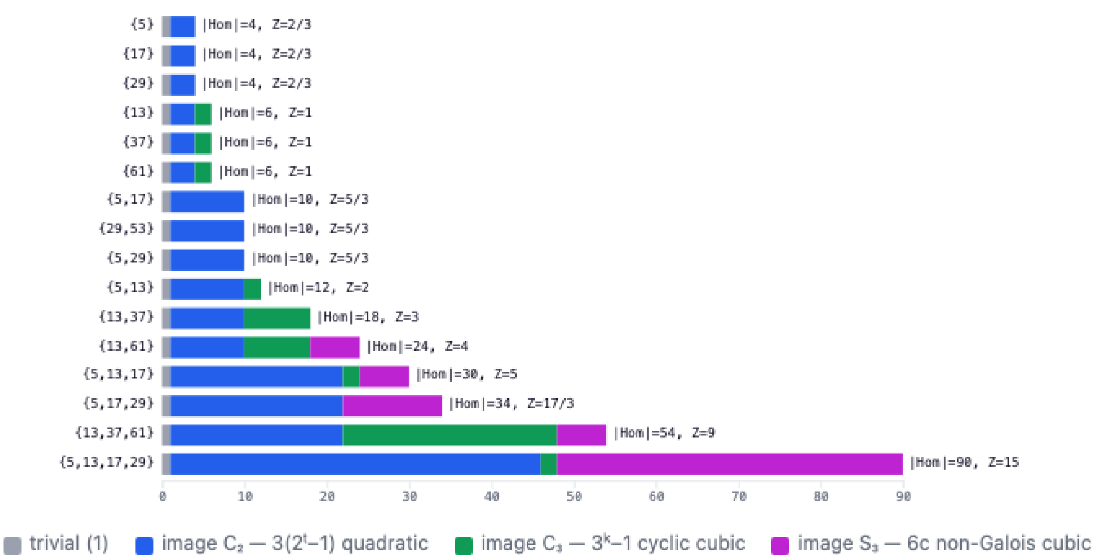
ACT II — STAGE 6

Non-abelian S_3 counting

The S_3 Dijkgraaf–Witten invariant of Spec $\mathbb{Z}[1/S]$

Stage 6 — Z_DW = $|\mathrm{Hom}(\pi_1^{et}, S_3)|/6$, decomposed by image: $1 + 3(2^t-1) + (3^k-1) + 6c$. The trivial/$C_2$/$C_3$ terms are our own machinery; the S_3 term 6c is refereed by cubic-field enumeration. Renders emitted JSON only.

$|\mathrm{Hom}(\pi_1, S_3)|$ decomposed by image, per prime set S



N_S_3 (= c, cubic iso classes) vs number of ramified primes t

`Z_DW` = $|\mathrm{Hom}(\pi_1, S_3)|/6 = 1/6 + (2^t-1)/2 + (3^k-1)/6 + c$. The S_3 term c (cubic iso classes) refereed **two independent ways**: PARI `nflist` and K. Belabas's `cubic`.

✓ verified · mass formula exact over 16 sweep instances (exact rational)

✓ verified · Belabas = PARI on all 14 feasible sets (2 skipped: $D_{max} = \prod p^2 >$ the 10^7 full-enumeration bound)

ACT III

The harness

The part that makes agreement mean something.

ACT III — THE FIVE CATEGORIES

How every claim is tested

- `twin_` — two independent algorithms agree (never optimize both; keep the naive one as referee).
- `theorem_` — a named theorem over an exhaustive/sampled range.
- `anchor_` — a specific published value (e.g. [13,61,937]=−1).
- `invariance_` — independence from an arbitrary choice (sqrt branch, solution, ordering).
- `oracle_` — PARI/GP, the Odlyzko table, Belabas's cubic — referee, never in `src/`; SKIP *visibly* when absent.

invariance_ is senior to anchor_. A passing anchor with a failing invariance means the convention is still wrong.

ACT III — PROCESS

Gates, pinning, pre-registration

- **Two-phase gates:** convention-free code first; the normalization is *pinned to a source* in `docs/notes/` before the load-bearing code is written.
- **Pre-registration:** Stage 4's LHS and RHS were built in separate translation units, in disjoint sessions, each seeing only its own spec. A mismatch would have been a *deliverable*, not a bug to reconcile.
- **Stratify the sweep — and sometimes prove the stratum void:** mod-8 third slot, corank cells, all-odd×odd-r (empty by parity), and the signature stratum *void by Stickelberger*.

ACT III — THE CENTERPIECE

ERRATA — everyone was caught

Every party that touched this project was caught in at least one error. Every catch was a computation or a citation — never an argument from authority.

```
roadmap generator — O_x, by a coherence check
spec author — two RESEARCH.md errata, by pinning + a witness value
human reviewer — R1, equal-parity (killed by the √65 witness), signature-mix (void by Stickelberger)
external (LLM) referee — a fabricated citation, by the verbatim-quote protocol — the same referee made a genuine catch
coding agent — strat counter (parity), signed-a (associativity), generator cap (sweep), 2c/6c (mass formula), PDFs-in-history, missing headers (a second compiler)
```

(A selection — the full ledger has 17 rows, including this deck review's own two.)

```
✓ verified · 17 entries in docs/ERRATA.md (selection above); authority-catches: 0
```

ACT IV

Results
and horizons

ACT IV — RESULTS

What was verified

- The full ladder, Stages 0–6, green.
- Every number on these slides is verified numerically over N cases, N pulled from emitted JSON or test output — never "proved" (that word is for literature theorems, with citation).
- Empirics are labeled observed, not asserted: Borromean rate 51.6%; negative-Pell fraction 0.666 vs the Koymans–Pagano asymptotic 0.5805; N_{S_3} growth.

✓ verified · 63 tests, 0 failing (10 oracle-refereed); extended run all green

ACT IV — HONESTY

What is NOT claimed

- Verified numerically over N cases ≠ proved. No slide asserts anything the suite doesn't test.
- Deninger's dynamical system is not constructed — the gap is the program, not a result.
- The mod-2 Chern–Simons case is degenerate (F_2-linear); the odd-p Gauss sum is future work.

ACT IV — HORIZONS

Where it goes

- **Odd-p arithmetic Chern–Simons** over totally imaginary fields (CKKPPY in-hypothesis) — the non-degenerate case, as future infrastructure.
- **Murmurations** — the field's live frontier: statistical structure in arithmetic data that this kind of harness is built to probe honestly.

References: RESEARCH.md §10. The pictures on these slides are the repo's own viewers, rendering the same emitted JSON the tests consume.